

Path-spread-resolved quantum transition-state theory for proton transfer

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Centroid quantum transition-state theory and ring-polymer molecular dynamics yield thermal rates, but they collapse the imaginary-time path of the transferring proton onto a single collective coordinate. We define the proton-path spread χ_p as the mean-square bead displacement of the ring polymer along the proton-sharing coordinate—the squared radius of gyration projected onto that coordinate—and use it to resolve the centroid dividing-surface ensemble. The associated path-spread-resolved (QRR) formula decomposes the usual centroid flux. When the transmission coefficient does not depend on path class, that integral is identically the centroid QTST rate k_{cTST} . For a harmonic proton mode, $\langle\chi_p\rangle$ has a closed continuum formula and an exact finite-bead counterpart; Lévy-staging path-integral Monte Carlo reproduces the finite-bead result to within 1%. The H/D ratio interpolates between $\sqrt{m_D/m_H}$ as $T \rightarrow 0$ and m_D/m_H as $T \rightarrow \infty$. Exact Eckart rates at 300 K give a tunneling correction $\kappa_H = 1.28\text{--}2.68$ and a kinetic isotope effect of 1.68–2.28 as the barrier rises from 4 to 16 $k_B T$ (classical isotope factor $\sqrt{2}$). Umbrella PIMC centroid QTST, which is QRR at $\kappa = 1$, overestimates those exact rates by 13–40% on converged points while remaining above TST; the Wigner series undershoots for H. The same dividing-surface ensemble has $f_{\text{deloc}}^H \approx 0.22\text{--}0.38$ versus $f_{\text{deloc}}^D \approx 0.02$. An asymmetric double well shows the same isotope contrast (0.18–0.23 versus 0.01–0.04). We do not compute a class-conditional recrossing factor. The estimator can be accumulated in any existing path-integral code.

Keywords: proton transfer, path-integral molecular dynamics, quantum transition-state theory, kinetic isotope effect, ring-polymer molecular dynamics

I. INTRODUCTION

Proton transfer, and proton-coupled electron transfer (PCET) in particular, is a setting in which nuclear quantum effects, solvent reorganization, and, at electrodes, the applied potential jointly control the rate.^{1–4} Kinetic isotope effects (KIEs) and non-Arrhenius temperature dependence are the usual experimental signatures of tunneling,^{4,5} but they do not by themselves identify the imaginary-time path geometry that carries the reactive flux.

Path-integral methods address that geometry at the level of the thermal density matrix.^{6–8} Centroid quantum transition-state theory (QTST) replaces the classical reaction coordinate by the ring-polymer centroid and evaluates a flux through a dividing surface in centroid space.^{9–12} Ring-polymer molecular dynamics (RPMD) propagates the same polymer in real time and supplies a recrossing correction to the centroid flux.^{13–15} Instanton theory, in the complementary deep-tunneling limit, optimizes a single dominant imaginary-time orbit.^{16,17} Transition-path sampling retains full real-time path information, at a cost that is not organized by equilibrium imaginary-time geometry.¹⁸

None of these approaches provides a cheap equilibrium order parameter that distinguishes compact ring poly-

mers (thermally activated, localized near the donor or acceptor) from delocalized polymers whose beads span the transfer coordinate (tunneling-like imaginary-time geometry). The ring-polymer radius of gyration is already used as a diagnostic of delocalization in condensed-phase proton studies.^{19,20} Here we restrict that diagnostic to the chemical sharing coordinate, give it exact harmonic limits, and use it to *condition* the QTST flux.

The resulting path-spread-resolved rate, which we call QRR, is

$$k_{\text{QRR}} = \int_0^\infty d\chi_p k_{\text{QTST}}(\chi_p) \kappa(\chi_p) p^\ddagger(\chi_p), \quad (1)$$

where $p^\ddagger(\chi_p)$ is the distribution of path spreads at the dividing surface and $\kappa(\chi_p)$ is a class-conditional transmission coefficient. When κ is taken independent of χ_p , Eq. (1) reduces identically to ordinary centroid QTST: $k_{\text{QRR}} = k_{\text{cTST}}$. QRR decomposes a known estimator rather than claiming a more accurate thermal rate. It reports which imaginary-time geometries carry the flux, how that mix depends on isotope and barrier, and whether path spread couples to a solvent coordinate.

Section II defines χ_p and the conditional free energy, states the solvable limits, and writes the flux decomposition. Section III describes the samplers. Section IV reports harmonic benchmarks, compares k_{cTST} (QRR at $\kappa = 1$) to exact Eckart rates, and resolves the dividing-surface ensemble of an asymmetric double well. Section V discusses scope and limitations. We do not report condensed-phase electrochemical simulations; those

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require a validated reactive potential and are left as an application of the estimator.

II. THEORY

A. Path-spread functional

The P -bead ring-polymer representation of the canonical density is standard.^{8,10} For a proton transferring between donor and acceptor oxygens we use the bead-resolved sharing coordinate

$$\xi_p^{(s)} = r_{\text{ODH}}^{(s)} - r_{\text{OAH}}^{(s)}, \quad \bar{\xi}_p = P^{-1} \sum_{s=1}^P \xi_p^{(s)}. \quad (2)$$

The path-spread functional is the bead variance along that coordinate,

$$\chi_p = \frac{1}{P} \sum_{s=1}^P (\xi_p^{(s)} - \bar{\xi}_p)^2. \quad (3)$$

This is the squared radius of gyration of the polymer projected onto ξ_p . It is identically zero at $P = 1$ and is non-negative for all configurations. Compact paths have χ_p of order the zero-point spread in a well; delocalized paths have χ_p large enough that beads occupy both sides of the barrier.

The joint density $P(\bar{\xi}_p, \chi_p)$ defines the conditional free energy

$$F(\bar{\xi}_p, \chi_p) = -k_B T \ln P(\bar{\xi}_p, \chi_p) + C. \quad (4)$$

Projection onto $\bar{\xi}_p$ recovers the centroid potential of mean force.¹⁰ Slices at fixed $\bar{\xi}_p$ resolve path class, as sketched in Fig. 1. A solvent or energy-gap coordinate λ_s may be added without changing the construction; the three-dimensional surface $F(\bar{\xi}_p, \chi_p, \lambda_s)$ is the natural object for PCET, but the calculations below are two-dimensional unless stated.

B. Flux decomposition

Let $\bar{\xi}_p^\ddagger$ locate the dividing surface. The marginal

$$p^\ddagger(\chi_p) = \frac{e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p)}}{\int d\chi_p' e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p')}} \quad (5)$$

is the probability that a dividing-surface configuration belongs to path class χ_p . The class-conditional QTST flux is the usual positive-velocity centroid flux restricted to that class,

$$k_{\text{QTST}}(\chi_p) = Z_R^{-1} \langle [\dot{\xi}_p] + \delta(\bar{\xi}_p - \bar{\xi}_p^\ddagger) \delta(\chi_p[\mathbf{R}] - \chi_p) \rangle. \quad (6)$$

A Bennett–Chandler transmission coefficient $\kappa(\chi_p)$ may be attached to each class.^{14,21,22} Integrating over χ_p produces Eq. (1). When κ does not depend on χ_p , the integral collapses to $k_{\text{QRR}} = \kappa k_{\text{QTST}}$, and the choice $\kappa = 1$ gives

$$k_{\text{QRR}}(\kappa = 1) \equiv k_{\text{cTST}}. \quad (7)$$

The Eckart calculations below evaluate this identity by computing k_{cTST} from umbrella PIMC and comparing it to the exact scattering rate. Two further restrictions apply. A Wigner series in $\hbar^2$ is a tunneling correction to a parabolic barrier,^{23,24} not a recrossing factor; the two must not be conflated. A χ_p -dependent κ is not computed: for highly delocalized paths the Wigner series is not a controlled approximation, and a class-conditional RPMD flux-side correlation would be the consistent dynamical correction.¹⁴ Constructing a $\kappa(\chi_p)$ from the centroid potential of mean force would double-count the barrier and cannot repair the QTST overestimate documented below.

C. Solvable limits

Remark 1 (Classical limit) At $P = 1$, Eq. (3) gives $\chi_p = 0$ identically, so $p^\ddagger(\chi_p) = \delta(\chi_p)$ and k_{QRR} reduces to classical transition-state theory. This is a consistency check, not an independent result.

Theorem 1 (Harmonic continuum) A harmonic mode $V = \frac{1}{2}m\omega^2 q^2$ has

$$\langle \chi_p \rangle_\infty = \frac{\hbar}{2m\omega} \coth\left(\frac{\beta\hbar\omega}{2}\right) - \frac{k_B T}{m\omega^2}. \quad (8)$$

The first term is the equal-time imaginary-time correlator $G(0)$; the second is the centroid variance, which is classical for a harmonic oscillator.¹⁰ The difference is non-negative because $\coth x \geq 1/x$ for $x > 0$. As $T \rightarrow 0$, $\langle \chi_p \rangle \rightarrow \hbar/(2m\omega)$. As $T \rightarrow \infty$, the leading term is the Wigner correction $\hbar^2\beta/(12m)$, independent of ω .²³ The Matsubara-mode derivation is given in the supplementary material.

Proposition 1 (Exact finite P) For the same oscillator discretized with P beads,

$$\langle \chi_p \rangle_P = \frac{1}{P} \sum_{k=1}^{P-1} \frac{1}{\beta\lambda_k}, \quad \lambda_k = \frac{4mP}{\hbar^2\beta^2} \sin^2\left(\frac{\pi k}{P}\right) + \frac{m\omega^2}{P}. \quad (9)$$

As $P \rightarrow \infty$, $\langle \chi_p \rangle_P \rightarrow \langle \chi_p \rangle_\infty$.

Equation (9) is the quantity that a primitive path-integral estimator of Eq. (3) actually samples. Comparing Monte Carlo to Eq. (8) at modest P therefore mixes statistical error with Trotter error. We compare to Eq. (9) first.

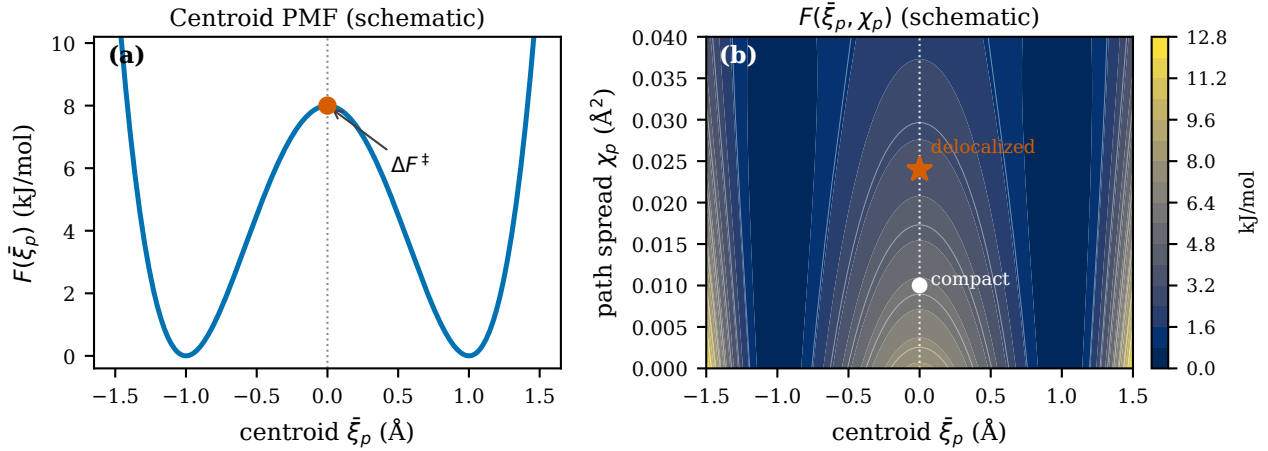


FIG. 1. Schematic, not computed data. (a) Centroid potential of mean force $F(\bar{\xi}_p)$, in which all ring-polymer shapes at a given centroid are averaged. (b) Conditional free energy $F(\bar{\xi}_p, \chi_p)$. Compact and delocalized dividing-surface geometries that share $\bar{\xi}_p = 0$ are marked.

Corollary 1 (Harmonic isotope scaling) *On a shared harmonic surface with $k = m\omega^2$ held fixed, $\langle \chi_p \rangle_H > \langle \chi_p \rangle_D$ at every $T > 0$. The ratio $\mathcal{R} = \langle \chi_p \rangle_H / \langle \chi_p \rangle_D$ satisfies*

$$\mathcal{R}(T \rightarrow 0) = \sqrt{m_D/m_H}, \quad \mathcal{R}(T \rightarrow \infty) = m_D/m_H. \quad (10)$$

The low-temperature limit is the ratio of zero-point amplitudes. The high-temperature limit is the ratio of leading Wigner corrections. Both follow from Eq. (8) and do not extend automatically to anharmonic barriers.

Proposition 2 (Factorization) *If $F(\bar{\xi}_p, \chi_p, \lambda_s) = F_p(\bar{\xi}_p, \chi_p) + F_s(\lambda_s)$, then $p^\ddagger(\chi_p)$ is independent of λ_s and the χ_p integral in Eq. (1) reduces to ordinary QTST when κ is class-independent.*

A nonzero mixed derivative $\partial^2 F / \partial \chi_p \partial \lambda_s$, or a nonzero dividing-surface correlation $\text{corr}(\chi_p, \lambda_s)$, would then signal proton-solvent coupling beyond a product Marcus picture.^{2,3} We do not claim to have measured that correlation with statistical control in this work.

III. METHODS

Path-integral Monte Carlo uses Lévy staging of the free-particle bridge,^{25,26} which samples the ring-polymer springs exactly, together with a Metropolis decision on the physical potential and a rigid centroid move. The primitive estimator of χ_p is the bead variance. For the harmonic oscillator we also evaluate Eqs. (8) and (9) in closed form.

Asymmetric double-well (ADW) free energies are obtained from one-dimensional umbrella sampling along $\bar{\xi}_p$ ²⁷ with harmonic restraints, reweighted by the

weighted histogram analysis method (WHAM).²⁸ Two-dimensional histograms of $(\bar{\xi}_p, \chi_p)$ use the same WHAM weights. The delocalized fraction at the dividing surface is

$$f_{\text{deloc}} = \int_{\chi_p \geq 2\chi_c} p^\ddagger(\chi_p) d\chi_p, \quad (11)$$

with the threshold $\chi_c = \hbar/(2m\omega^*)$ evaluated at $\omega^* = 1000 \text{ cm}^{-1}$, giving $\chi_c \approx 0.010 \text{ Å}^2$. Compact and intermediate classes use $0.5\chi_c$ and $2\chi_c$ as bookkeeping cuts; the continuous $p^\ddagger(\chi_p)$ is the primary object.

Exact thermal rates for the symmetric Eckart barrier $V(x) = V_0 \text{sech}^2(x/a)$ are

$$k = \frac{1}{2\pi\hbar Q_r} \int_0^\infty dE e^{-\beta E} T(E), \quad (12)$$

with $T(E)$ the exact transmission probability of the symmetric Eckart barrier²⁹ and $Q_r = (m/2\pi\hbar^2\beta)^{1/2}$. Classical TST is $k_{\text{TST}} = \sqrt{k_B T / (2\pi m)} e^{-\beta V_0}$. The Wigner estimate multiplies k_{TST} by $1 + (\beta\hbar\omega_b)^2/24$, where $\omega_b = \sqrt{2V_0/(ma^2)}$. All Eckart integrals use a logarithmically stable evaluation of $T(E)$.²⁹

Centroid QTST on the same barrier is obtained from umbrella PIMC along the centroid,

$$k_{\text{cTST}} = \sqrt{\frac{k_B T}{2\pi m}} e^{-\beta F(0)}, \quad (13)$$

with F the WHAM potential of mean force, zeroed on the mean of the $|\bar{\xi}_p| > 1.6 \text{ Å}$ tails so that $F(\infty) = 0$. Eleven harmonic windows on $[-2, 2] \text{ Å}$ with $k_{\text{umb}} = 60 \text{ kJ mol}^{-1} \text{ Å}^{-2}$ are used, $P = 32$ (H) and $P = 24$ (D), 4×10^4 production steps after 8×10^3 equilibration. Residual tail scatter in F is of order 0.5 kJ mol^{-1} . Deuterium at $V_0 = 12 k_B T$ is excluded from the rate comparison:

the dividing-surface window drifted and produced the unphysical ordering $F_D(0) < F_H(0)$. With $\kappa = 1$, Eq. (1) equals Eq. (13) identically; both values are stored in the accompanying JSON.

Temperatures are 300 K unless noted. Masses are 1 and 2 amu. Random number generators are seeded at 42. Code and JSON outputs are in the repository accompanying this article.

IV. RESULTS

A. Harmonic oscillator

Figure 2 compares Eq. (8) with Lévy-staging Monte Carlo for a 1000 cm^{-1} mode over 100–700 K. At 300 K the continuum values are $\langle \chi_p \rangle_H = 0.01011 \text{ \AA}^2$ and $\langle \chi_p \rangle_D = 0.00572 \text{ \AA}^2$. With $P = 64$ (H) and $P = 32$ (D), the exact finite-bead formula differs from the continuum by 0.12% and 0.29%, respectively, and the Monte Carlo root-mean-square relative error against Eq. (9) is 0.56% (H) and 1.03% (D).

The isotope ratio $\mathcal{R}(T)$ increases from 1.52 at 100 K toward $\sqrt{2}$ as T is lowered, and from 1.77 at 300 K to 1.94 at 700 K, approaching 2 from below, as required by Corollary 1 [Fig. 2(b)]. Figure 2(c) shows that the primitive estimator tracks Eq. (9) from $P = 4$ to $P = 128$ and that the Trotter error relative to the continuum is already below 0.5% at $P = 32$ for this frequency and temperature. A staging move that draws Brownian-bridge *marginals* independently, rather than the sequential Lévy construction, produces a spurious increase of $\langle \chi_p \rangle$ with P and should not be used.

B. Eckart barrier

The symmetric Eckart barrier with width $a = 0.50 \text{ \AA}$ supplies exact thermal rates that do not depend on path-integral sampling, and therefore a test of Eq. (7): k_{QRR} at $\kappa = 1$ is k_{cTST} , and that number can be compared with the exact scattering rate.

Figure 3(a) and Table I compare four estimates of $\kappa = k/k_{\text{TST}}$ at 300 K. The exact κ_H increases from 1.28 at $V_0 = 4 k_{\text{B}}T$ to 2.68 at $V_0 = 16 k_{\text{B}}T$; the corresponding Wigner factors are 1.22 and 1.86. Wigner is a useful first correction at moderate $\hbar\omega_b/k_{\text{B}}T$ and underestimates the exact κ once $\hbar\omega_b/k_{\text{B}}T \gtrsim 4$. Centroid QTST from umbrella PIMC gives $\kappa_H = 1.54, 1.98$, and 2.30 at 4, 8, and $12 k_{\text{B}}T$, versus exact 1.28, 1.60, and 2.04 (overestimates of 20%, 24%, and 13%). Deuterium is overestimated by 40% and 10% at 4 and $8 k_{\text{B}}T$. In every converged case $F(0) < V_0$, so $\kappa_{\text{cTST}} > 1$ and the nuclear quantum effect has the correct sign. Deuterium at $12 k_{\text{B}}T$ is omitted: the dividing-surface umbrella drifted and produced the unphysical ordering $F_D(0) < F_H(0)$.

The exact KIE [Fig. 3(b)] is 1.68 at $4 k_{\text{B}}T$ and 2.28

at $16 k_{\text{B}}T$, against a classical factor $\sqrt{2} \approx 1.414$. Centroid QTST gives 1.44 and 1.98 at 4 and $8 k_{\text{B}}T$: it barely exceeds classical at the lowest barrier, where H and D overshoot by similar factors, and it overshoots the exact KIE at $8 k_{\text{B}}T$. These comparisons are the numerical content of Eq. (1) at $\kappa = 1$ and do not imply that QRR improves on QTST.

The integrated rate alone does not resolve this isotope contrast; Fig. 3(c) shows it directly. At the dividing surface, $f_{\text{deloc}}^H = 0.22, 0.23$, and 0.38 at 4, 8, and $12 k_{\text{B}}T$, while f_{deloc}^D remains 0.017. The jump at $12 k_{\text{B}}T$ for H is sampled with 629 dividing-surface configurations and should be read as a trend, not a precise fraction. The isotope contrast of an order of magnitude is the robust result.

C. Asymmetric double well

The ADW $V(q) = V_0[(q/q_0)^2 - 1]^2 + \alpha q$ with $q_0 = 0.70 \text{ \AA}$ is a bound proton-transfer model. Figure 4(a) shows the WHAM centroid PMF at $V_0 = 8 k_{\text{B}}T$: the H barrier is 15.9 kJ mol^{-1} and the D barrier is 17.6 kJ mol^{-1} . Figure 4(b) is the quantity QRR actually uses, the dividing-surface density $p^\ddagger(\chi_p)$. It is broader for H than for D, with means $\langle \chi_p \rangle_\ddagger^H = 0.015 \text{ \AA}^2$ and $\langle \chi_p \rangle_\ddagger^D = 0.008 \text{ \AA}^2$. With the bookkeeping cut $2\chi_c = 0.020 \text{ \AA}^2$, the delocalized weight is $f_{\text{deloc}}^H = 0.18\text{--}0.23$ across $V_0 = 2\text{--}8 k_{\text{B}}T$, versus $f_{\text{deloc}}^D = 0.014\text{--}0.045$ [Fig. 4(c) and Table II]. For H the fraction is essentially flat in this barrier window; the robust observation is the isotope contrast, not a strong barrier trend.

These are equilibrium path-class statistics. The isolated ADW is a bound well and has no irreversible thermal rate, so we do not convert f_{deloc} into a QRR rate, nor into a claimed improvement over RPMD. Two-dimensional histograms of $F(\xi_p, \chi_p)$ from the same umbrellas are too sparsely sampled in χ_p to publish as free-energy surfaces.

D. What is not shown

A bilinear proton-solvent model $V = V_{\text{ADW}}(q_p) + \frac{1}{2}k_s q_s^2 + \kappa_c q_p q_s$ was sampled without umbrellas. Dividing-surface correlations $\text{corr}(\chi_p, q_s)$ lie between -0.06 and -0.16 with a few hundred crossings and do not trend cleanly with κ_c . That scan is not evidence for or against Proposition 2; a controlled test would require umbrella sampling in $(\xi_p, \chi_p, \lambda_s)$. We therefore omit it from the figures.

V. DISCUSSION

QRR does not replace RPMD, instanton theory, or vibronic PCET rate laws.^{2,14,17,30} RPMD remains the ap-

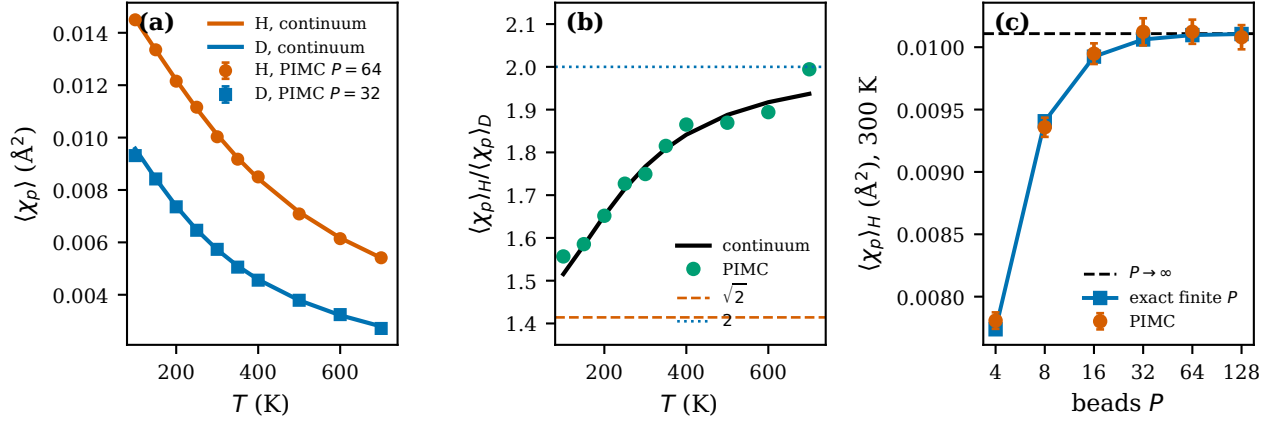


FIG. 2. Harmonic path spread for $\omega = 1000 \text{ cm}^{-1}$. (a) $\langle \chi_p \rangle(T)$ from Eq. (8) (lines) and Lévy-staging PIMC (symbols). (b) Isotope ratio, with the $T \rightarrow 0$ limit $\sqrt{2}$ and the $T \rightarrow \infty$ limit 2. (c) Bead convergence of $\langle \chi_p \rangle_H$ at 300 K: PIMC versus the exact finite- P formula, Eq. (9).

TABLE I. Symmetric Eckart barrier at $T = 300 \text{ K}$, $a = 0.50 \text{ \AA}$. $\kappa = k/k_{\text{TST}}$. Centroid QTST is umbrella PIMC; QRR at $\kappa = 1$ equals that column. Dashes mark points that were not computed, or D at $12 k_B T$, which failed the dividing-surface check. Classical KIE is $\sqrt{2}$ at every barrier.

$V_0/k_B T$	κ_H^{ex}	κ_H^{W}	κ_H^{cTST}	κ_D^{ex}	κ_D^{W}	κ_D^{cTST}	KIE ^{ex}	KIE ^W	KIE ^{cTST}
4	1.28	1.22	1.54	1.08	1.11	1.51	1.68	1.55	1.44
8	1.60	1.43	1.98	1.28	1.22	1.41	1.77	1.67	1.98
12	2.04	1.65	2.30	1.36	1.32	—	2.12	1.76	—
16	2.68	1.86	—	1.66	1.43	—	2.28	1.84	—

TABLE II. Asymmetric double well at $T = 300 \text{ K}$. Centroid barriers from 1D WHAM; f_{deloc} from the WHAM-weighted dividing-surface ensemble with $2\chi_c = 0.020 \text{ \AA}^2$.

$V_0/k_B T$	ΔF_H (kJ/mol)	ΔF_D (kJ/mol)	f_{deloc}^H	f_{deloc}^D	$\langle \chi_p \rangle_{\ddagger}^H$ ($\text{\AA}^2$)	$\langle \chi_p \rangle_{\ddagger}^D$ ($\text{\AA}^2$)
2	4.25	4.60	0.18	0.014	0.014	0.007
4	8.90	9.02	0.23	0.020	0.015	0.007
6	12.4	14.3	0.22	0.024	0.015	0.007
8	15.9	17.6	0.23	0.045	0.015	0.008

appropriate dynamical correction to centroid QTST in the crossover regime; instanton theory remains the appropriate semiclassical description when a single tunneling orbit dominates; Hammes-Schiffer vibronic theory remains the appropriate nonadiabatic framework when electron-proton nonadiabaticity is essential. χ_p adds an equilibrium resolution of imaginary-time geometries that those methods otherwise average or optimize.

The present calculations evaluate Eq. (1) only at $\kappa = 1$, where it is identical to k_{cTST} . A class-conditional transmission coefficient is not computed. On the Eckart barrier that common rate overestimates the exact scattering result by 13–40% on converged points, as expected for a recrossing-free centroid theory; QRR cannot reduce that error without a genuine $\kappa(\chi_p)$. The ADW section reports

$p^\ddagger(\chi_p)$ only. Connecting the two models is an analogy: the isolated double well has no irreversible thermal rate.

Two condensed-phase hypotheses can be drawn from these results, each requiring a separate test. If an increase of the observed KIE with driving force^{30,31} is carried by tunneling-like paths, $\Delta f_{\text{deloc}} = f_{\text{deloc}}^H - f_{\text{deloc}}^D$ should increase with that driving force. A breakdown of $F = F_p + F_s$ at the dividing surface would signal proton-solvent coupling beyond a product Marcus picture. Neither is tested here.

χ_p is an imaginary-time polymer statistic, not a spectroscopic observable. The cut χ_c is bookkeeping; Fig. 4(b) is the primary object. Recrossing is not resolved by class. The models are one-dimensional. Residual WHAM tail scatter of order 0.5 kJ mol^{-1} limits the precision of κ_{cTST} at the lowest Eckart barrier. Finite- P errors in χ_p are under control for the 1000 cm^{-1} mode at $P \geq 32$; stiffer stretches will need more beads or a virial estimator.

VI. CONCLUSIONS

We have defined a projected ring-polymer path spread χ_p for proton transfer, derived its harmonic continuum and finite-bead limits, and used it to resolve the centroid dividing-surface ensemble. Lévy-staging Monte Carlo re-

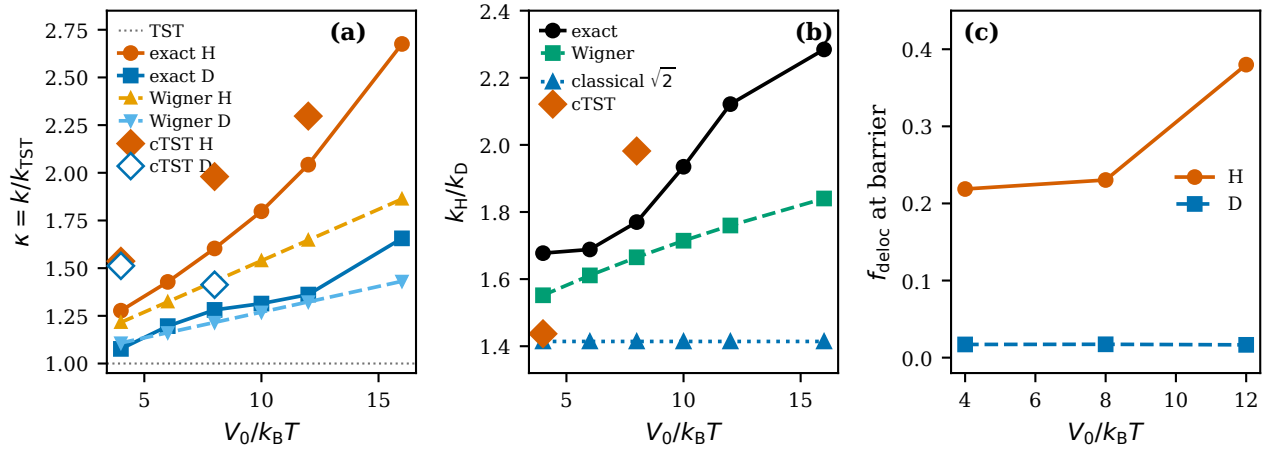


FIG. 3. Symmetric Eckart barrier, $a = 0.50 \text{ \AA}$, $T = 300 \text{ K}$. (a) Tunneling correction $\kappa = k/k_{\text{TST}}$. Diamonds are umbrella-PIMC centroid QTST, equal to QRR at $\kappa = 1$. Deuterium at $12 k_B T$ is omitted (dividing-surface window not converged). (b) Kinetic isotope effect. Classical TST is $\sqrt{2}$ at every V_0 . (c) Delocalized weight at the dividing surface, $2\chi_c = 0.020 \text{ \AA}^2$.

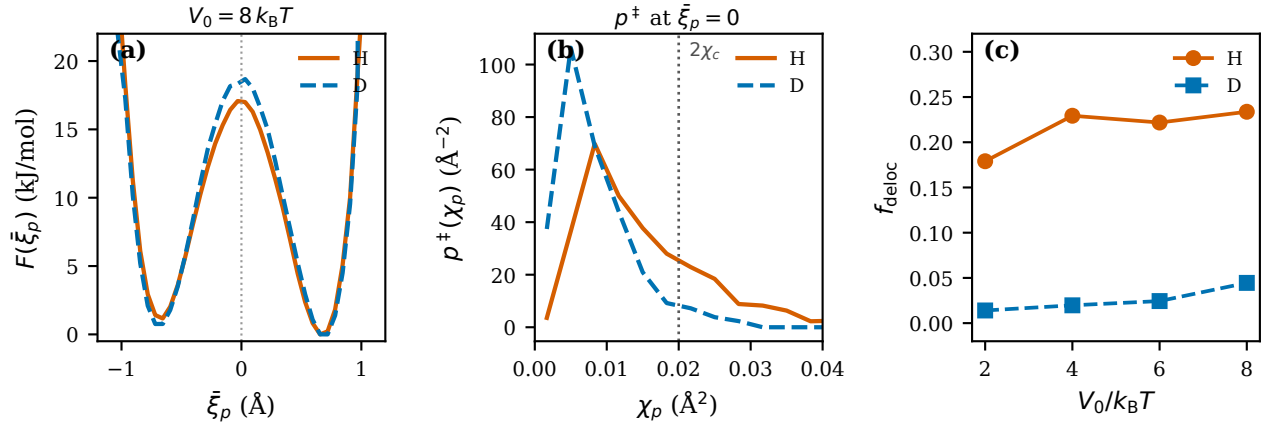


FIG. 4. Asymmetric double well, $T = 300 \text{ K}$. (a) Centroid PMF at $V_0 = 8 k_B T$. (b) Dividing-surface path-spread density $p^\dagger(\chi_p)$ at the same barrier; the dotted line is the bookkeeping cut $2\chi_c$. (c) Delocalized fraction versus barrier height. The H fraction is nearly flat; the D fraction remains an order of magnitude smaller.

produces the exact finite-bead harmonic formula to about one percent. Exact Eckart rates quantify when tunneling corrections and KIEs depart from TST. Umbrella PIMC centroid QTST, equal to QRR at $\kappa = 1$, overestimates those exact rates while capturing the sign of the nuclear quantum effect and assigning a much larger delocalized weight to H than to D. On an asymmetric double well the dividing-surface path-spread distribution is robustly broader for H than for D. Equation (1) decomposes QTST and is not claimed to produce a more accurate rate. The estimator is a collective variable and can be accumulated in any path-integral molecular dynamics run.

ACKNOWLEDGMENTS

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.22718950> (Ref. 32). The source repository is <https://github.com/vmercel/QRR-PCET>.

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